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Expanding scope of genetic studies in the era of Biobanks

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Abstract

Biobanks have become pivotal in genetic research, particularly through genome-wide association studies (GWAS), driving transformative insights into the genetic basis of complex diseases and traits through the integration of genetic data with phenotypic, environmental, family history, and behavioral information. This review explores the distinct design and utility of different biobanks, highlighting their unique contributions to genetic research. We further discuss the utility and methodological advances in combining data from disease-specific study or consortia with that of biobanks, especially focusing on summary statistics based meta-analysis. Subsequently we review the spectrum of additional advantages offered by biobanks in genetic studies in representing population differences, calibration of polygenic scores, assessment of pleiotropy and improving post-GWAS in silico analyses. Advances in sequencing technologies, particularly whole-exome and whole-genome sequencing, have further enabled the discovery of rare variants at biobank scale. Among recent developments, the integration of large-scale multi-omics data especially proteomics and metabolomics, within biobanks provides deeper insights into disease mechanisms and regulatory pathways. Despite challenges like ascertainment strategies and phenotypic misclassification, biobanks continue to evolve, driving methodological innovation and enabling precision medicine. We highlight the contributions of biobanks to genetic research, their growing integration with multi-omics, and finally discuss their future potential for advancing healthcare and therapeutic development.

Introduction.

Genome-wide association studies (GWAS) have revolutionized our understanding of the genetic basis of complex diseases[1]. Over the past decade, advancements in genotyping technologies, high throughput sequencing, coupled with advances in statistical computational techniques have exponentially increased the utility of GWAS[2–4]. This has further been fueled by increasing deep phenotyping capabilities through imaging, advanced biochemical assays, wearable devices, longitudinal assessments, and others, extending our understanding of traits and diseases to more granular levels, beyond standard clinical outcomes. In recent times, inclusion of diverse populations extending to underrepresented

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populations and improving global generalizability have expanded both the applicability and methodological innovations in GWAS[5,6]. Furthermore, post-GWAS in silico analyses as well as integration with functional genomics and multi-omics data has enhanced our understanding of disease related pathways and mechanisms, propelling precision medicine and therapeutic innovation[7–9].

The primary driver of the growth of the scale and applicability of GWAS in the last decade has been the establishment and expansion of large biobanks[10] which have steadily encompassed majority of the aspects described above, including high throughput genotyping, genome-sequencing, deep phenotyping and population differences. These resources combine genetic data with health records, behavioral, family history, environmental and lifestyle information for a large number of individuals, providing a unique and extensive platform for advanced genetic studies. A notable aspect is the deep and rich phenotypic data often collected in biobanks, increasingly using state-of-the-art technology, which have advanced our understanding of the interplay between genetic and environmental susceptibility factors[11,12], disease progression and treatment outcomes. Further, with sequencing technologies, biobanks offer an unprecedented opportunity of cataloging the effects of rare genetic variants[13,14]. More recently, population heterogeneity focused biobanks have aimed at uncovering ancestry-specific variants that influence traits and diseases[13,15].

This review explores the role of biobanks in GWAS, highlighting differences in their design, advantages of data integration, the incorporation of multi-omics, and the study of rare variants. We also discuss challenges and future opportunities for leveraging biobanks in genetic research.

Design and themes of biobanks.

Although introduced in 1996, there is still a lack of consensus on the precise definition of “biobank”[16,17]. In 2009, Organization for Economic Cooperation and Development (OECD) described biobanks as “structured resources that can be used for the purpose of genetic research and which include (a) human biological materials and/or information generated from the analysis of the same and (b) extensive associated information”[18]. Currently, the genetic research community leverages data from primarily two types of biobanks, which we discuss in detail below. However, with the evolving definition of biobanks and the ever-increasing scale of genetic research, such themes do not constitute an exhaustive list. In fact, we see several additional emerging themes which include larger biorepositories and/or biological resource centers.

Population-based biobanks.

Population-based biobanks capture genetic and health-related data from a wide cross-section of the general population, offering a broad representation of the spectrum of genetic and phenotypic differences. Early examples include the Icelandic Biobank (deCODE Genetics), established around 1996, which linked genetic data with genealogical and health records. Subsequently the UK Biobank (UKB; 2006)[19], Estonian Biobank (2000)[20], Biobank Japan (2003)[21], and China Kadoorie Biobank[22], FinnGen[23] and others have emerged, forming

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the foundation of region- and country-specific health research (Figure 1A). More recent initiatives like[23] H3Africa[24], Mexican Biobank[25] and others have prioritized regional genetic differences and study of underrepresented populations. A key strength of population-based biobanks is their scale, enhancing the statistical power of genetic studies and enabling the discovery of variants with small effect sizes for common diseases such as heart disease, diabetes, and mental health conditions. Integrating deep phenotypic data, family history, behavioral, and environmental data allows researchers to explore the interplay between genetic and environmental factors. Longitudinal data, as catalogued in the UKB, further aids in understanding disease progression over time[26–28]. However, results from studies in such biobanks can depend on the recruitment strategies resulting in overrepresentation of certain groups, such as healthier, or more motivated participants, reducing the generalizability to the broader population[29,30]. Additionally, while broad, phenotypic data often lack the depth needed for detailed studies of specific diseases.

EHR/Hospital-Based Biobanks.

Electronic health records (EHR)-based biobanks represent a transformative resource in genetic research, combining the richness of real-world medical data with genotypic and phenotypic information. These biobanks include hospital, health care provider network, or health system-based studies such as BioVU[31], Michigan Genomics Initiative[32], BioMe[33,34], Partners Biobank[35,36] and others[37–39], and more recently, the nation-wide All of Us Research Program (AoU)[15,40] and the Million Veterans Program[41] (Figure 1B). The primary advantage of EHR-based biobanks is their ability to provide large-scale, longitudinal clinical data by linking genetic data to comprehensive medical histories, enabling investigation of temporal variation in disease progression, treatment outcomes, and environmental interactions[42]. For example, BioVU has been instrumental in identifying genetic variants associated with drug response, supporting advances in pharmacogenomics[43–45]. The recently published Million Veteran Program highlights the benefits of including individuals from distinct populations[41]. Another key strength is the deep phenotypic data collected in terms of clinical variables, such as diagnoses, medications, lab results, and imaging data, providing a robust foundation genetic studies of multifactorial diseases. However, implicit overrepresentation of certain sub-populations due to the selection schemes, remains a significant challenge in EHR-based biobanks due to their design. While most population-based biobanks ascertain disease endpoints using population-based registries, EHR-based biobanks like AoU determine disease endpoints by baseline surveys and records from participating health care providers/systems. Hence a (near-) complete enumeration of incident and prevalent diseases is available for population-based biobanks while that in EHR-based biobanks is complicated as individuals may not participate in health systems, or they may move in and out of health systems. This generates non-random missing data and censoring, limiting the generalizability of studies based on EHR biobanks. Further, variability in clinical coding practices and the quality of EHR data can lead to phenotypic misclassification.

There are several additional themes of biobanks and enhanced biorepositories emerging which we list below. Again, this list is not exhaustive and will likely grow in the coming years.

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Birth cohort-based biobank.

These are specialized resources designed to study genetic and environmental influences on health from early life to adulthood, following individuals from birth, often starting with prenatal data, and collect extensive genetic, phenotypic, and environmental information over time. Examples include ALSPAC (UK)[46], MoBa (Norway)[47], and the Generation R Study (Netherlands)[48,49]. Their longitudinal design allows researchers to explore how early-life factors shape long-term health outcomes, aiding the study of developmental processes, chronic diseases and gene-environment interactions. However, such biobanks are resource-intensive, especially for follow-up and have limited applicability to adult-onset diseases.

Clinical-trial biobanks.

Clinical trial-based biobanks collect genetic and biological data from trial participants, linking genetic variation to treatment responses, adverse effects, and therapeutic efficacy. The key advantage is the high quality and specificity of the phenotypic data they provide due to advanced monitoring, with standardized data collection protocols that ensure consistency and reliability. For instance, the NCI-MATCH biobank[50] has successfully used tumor genetic alterations to predict responses to targeted cancer therapies, while the Biomarkers Consortium Biobank[51] has linked genetic variants to drug efficacy and toxicity. These biobanks also inform drug development by identifying patient subgroups likely to benefit from specific treatments, optimizing therapeutic outcomes but are often limited in scale.

Combining disease-based consortia and Biobanks.

Detecting associations between genetic variants and complex traits requires large sample sizes, particularly for common variants which usually have weaker effect sizes[52]. While the scale of large biobanks exemplify this advantage, this might not be sufficient for studying specific diseases. Often they lack adequate number of cases for lesser prevalent diseases (especially in the target population) reducing statistical power for GWAS[53,54]. In contrast, disease-specific studies or consortia are formed explicitly to study the genetics and biology of a particular disease or group of related conditions. This singular focus allows them to address key challenges in understanding disease etiology that biobanks, with their broader scope, may not be able to tackle as effectively.

There are interesting methodological challenges for effectively combining disease-related consortia and biobanks, leveraging their respective strengths for genetic studies of specific diseases. Typically disease-based consortia often have limited or no information on environmental and/or risk-factors thereby stifling downstream genetic epidemiological studies. In contrast, biobanks allow prospective analysis of association between risk-factors and future outcome providing opportunity to conduct more elaborate analysis involving risk factors, such as gene-environment interactions and mediation analysis. Combining the strengths of both designs—leveraging consortia's precision with biobanks' breadth—can enhance the impact of genetic research. In this respect, meta-analysis of disease-consortia data and multiple biobanks have become standard approach in current disease GWAS since it can be performed with only summary level data, circumventing various logistical, ethical and privacy concerns of accessing the individual level data required for pooled

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analysis[56–58]. However, standard meta-analysis can only combine information across studies through a common set of harmonized parameters and thus cannot really be used to combine information from more complex models which could be built based on biobanks and GWAS summary-statistics data which are typically built derived from very simplistic models with minimal adjustment for covariates. In this regard, modern data integration[55] and generalized meta-analysis[56] could provide opportunities for various novel analyses for combined analysis of biobanks and external GWAS data.

Additional aspects of biobanks.

Genetic studies in biobanks have multiple different facets as shown in Figure 2.

Methods for large-scale GWAS.

One of the most significant advancements in GWAS has been the dramatic increase in sample sizes, largely due to the growth of biobanks. However, the unprecedented scale of biobanks have triggered the development of computationally efficient methods of GWAS and quantifying genetic contributions to diseases. Even in the presence of cryptic relatedness within the study participants, advanced statistical methods and computational techniques have dramatically decreased the time needed for GWAS[57,58]. In addition, estimation of genetic architecture parameters like heritability and polygenicity have also seen methodological improvements to incorporate the scale of biobanks[52,59–61].

Population differences.

GWAS have primarily focused on European ancestry, limiting the applicability of findings to other populations. Initiatives like the AoU and H3Africa aim to study a broader spectrum of genetic variation, to uncover ancestry-specific variants and their functions. Additionally combining data from different biobanks, potentially covering different populations, has revealed differences in variant frequencies and effects linked to complex disease like cardiovascular disease, highlighting how such efforts improve our understanding of the differences in manifestation of disease traits across populations. For example, differences in frequency of genetic variants related to Glucose-6-phosphate dehydrogenase deficiency has been shown to partially explain the differential prevalence of diabetes-related complications across ancestral populations[62]. These efforts have also driven methodological advancements in multi-ancestry GWAS and post-GWAS pipelines[63–68], explicitly incorporating genetic ancestry to account for population heterogeneity. A currently popular school of thought is to consider genetic ancestry as a continuum rather than assigning participants into discrete groups, which is particularly attractive due to increasing participation of admixed individuals[69]. However, a comprehensive review of the statistical methods leveraging multi-ancestry and diverse population data is beyond the scope of the current work.

Development and calibration of polygenic risk scores (PRS).

PRS aggregate the effects of multiple genetic variants to estimate an individual’s genetic predisposition to specific traits or diseases. The extensive sample sizes of biobanks enable the detection of variants with weaker effect sizes that would otherwise be missed, potentially improving the predictive power of PRS. Coupled with the increasing focus on genetically

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heterogeneous study participants, multi-ancestry PRS methods have also been proposed and have shown promise in providing equitable risk estimates across multiple ancestries. However a key area of application of PRS for biobanks are in calibration[70]. Biobanks such as the UKB, FinnGen, and the Estonian Biobank link genetic information with comprehensive health records, including longitudinal disease incidence data. This linkage allows researchers to evaluate how well PRS predict actual disease incidence across populations. For instance, the UKB's large cohort of 500,000 participants has enabled the calibration of PRS for conditions like coronary artery disease, type 2 diabetes, and breast cancer by comparing predicted risks with observed disease rates[71,72]. Disease incidence data provide real-world benchmarks, ensuring that PRS are not only statistically robust but also clinically meaningful.

Assessment of pleiotropy.

Biobanks facilitate pleiotropy assessment by linking genetic data with extensive phenotypic datasets. Methods like Genomic SEM and LD Score Regression use biobank data to estimate genetic correlations and shared heritability[73,74], while multi-trait PRS approaches leverage pleiotropy to improve predictive accuracy[75,76]. Phenome-wide association studies (PheWAS) further advance pleiotropy research by associating genetic variants with a wide range of phenotypes[77,78]. Biobanks such as UKB, Vanderbilt BioVU, and FinnGen have driven PheWAS, linking genotypic data to a multitude of phenotypes which is further broadened by[83,84] AoU through ancestry-specific pleiotropy studies in underrepresented populations.

Post-GWAS studies: fine mapping and mendelian randomization:

Biobanks are critical for advancing post-GWAS analyses, such as fine mapping and Mendelian Randomization (MR). While GWAS identify regions of association, fine mapping pinpoints causal variants by resolving linkage disequilibrium (LD). Large biobank datasets improve LD estimation accuracy proving to be crucial for resolving complex LD patterns and pinpointing causal variants. Advent of multi-ancestry biobanks further improve the resolution fine mapping by leveraging population differences in LD structures and allele frequencies[65,79]. MR uses genetic variants as instruments to infer causal relationships between exposures and outcomes[80]. Biobanks provide extensive genetic and phenotypic data, supporting robust MR analyses[81,82]. Combining disease-specific summary statistics with population biobank data strengthens causal inference. Additionally, biobanks with longitudinal data facilitate time-sensitive MR studies, distinguishing between early-life and adult exposures[83]. This has provided valuable insights into the genetic basis of long-term health outcomes and preventive strategies.

Growing multi-omic studies in biobanks.

While external resources like GTEx[84], eQTLGen[85], and ENCODE[86] have pioneered omics integration with GWAS coupled with methodological advances[9,87–90], biobanks now increasingly generate their own multi-omics datasets, enabling deeper insights into causal pathways and molecular targets through joint analysis of phenotype, genotype and omics data. Proteomics, in particular, has gained traction due to advancements in cost-effective

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high-throughput assays like SomaScan and Olink. Several biobanks have conducted large-scale proteomics studies, most recently with the UKB's Pharma Proteomics Project profiling plasma proteins in 54,219 participants[91]. Proteogenomic analyses have identified genetic variants regulating circulating protein levels, with post-GWAS techniques like Mendelian Randomization uncovering disease mechanisms and biomarkers, including for cancer[92–94]. Additionally, in conjunction with the follow-up or incidence data on diseases, integrative proteomics can further be used to identify proteins whose dysregulation can potentially be disease-causing[95].

Among other omics, metabolomic panels, in particular, have been used by several biobanks given their target population and potentially increasing prevalence of metabolic disorders[96]. However data on transcriptomics, epigenomics, and metagenomics, have remained fairly limited due to higher cost. Beyond these, the UKB has also pioneered the genetic study of brain MRI and X-ray images[97] due to availability of deep phenotyping data. While multi-omic data generation remains resource-intensive, more recent initiatives like the AoU have devoted funds to profile and integrate diverse omics data to uncover ancestry-specific pathways and biomarkers, driving precision medicine and addressing health disparities.

Sequencing studies: Rare Variant Discovery and Association

Biobanks are pivotal for studying rare genetic variants, leveraging sequencing technologies like whole-exome sequencing (WES) and whole-genome sequencing (WGS). WES, focusing on protein-coding regions (1–2% of the genome), is cost-effective and ideal for identifying rare coding variants with large effect sizes, especially in Mendelian disorders or protein-coding regions linked to complex diseases[98]. In contrast, WGS captures the entire genome, allowing the discovery of noncoding variants, structural rearrangements, and regulatory elements influencing disease[1313,99,100].

While findings from these approaches often complement each other, the choice between WGS and WES often depends on the specific research goals, available resources, and the population being studied[101]. While WES is cost-effective, it excludes large portions of the genome, potentially missing important regulatory variants. In comparison, WGS, although comprehensive, is resource-intensive and may produce large amounts of data that are difficult to interpret or store. Till date, majority of the actionable findings from rare variant studies have largely been driven by WES while the discoveries from WGS have been modest. However, with the integration of multi-omic data in several larger biobanks, WGS holds potential to uncover the regulatory effects of rare variants. Conversely, sequencing diverse populations, as seen in the AoU[15], enhances the discovery of ancestry-specific rare variants but requires sophisticated methods to account for population stratification and linkage disequilibrium differences.

Discussions

In the last decade, biobanks have had a transformative impact for research on genetic basis of complex diseases and traits. Subsequent and more recent expansion into multi-omic data, coupled with methodological advances, have proven to pivotal resources that are actively

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being queried by researchers to develop newer insights into genetic architecture of diseases leading to improved risk prediction and identification of newer therapeutic targets. Next we list the potential challenges in using biobank data and outline the future opportunities in the forthcoming years.

Challenges and Pitfalls of GWAS in Biobanks.

Despite their benefits, biobank-based genetic studies also present challenges. One significant issue is the potential for implicit overrepresentation of certain individuals, as biobank participants are often volunteers who may not represent the general population[29,102]. For example, studies based on the UKB tend to underrepresent individuals from lower socioeconomic backgrounds and those with severe health conditions, potentially skewing results. Another challenge is phenotypic misclassification. While EHRs offer rich phenotypic data, they may contain inaccuracies or inconsistencies due to varying clinical coding practices. Additionally, ethical considerations related to privacy and informed consent are paramount in biobank research. The use of genetic data linked to EHRs requires robust governance structures to protect participants' privacy and manage data access.

Potential for Precision Medicine.

Looking forward, biobank-based GWAS has the potential to transform precision medicine. By identifying individuals with high polygenic risk scores, healthcare providers could offer preventive interventions tailored to a patient's genetic risk profile. For instance, people with high PRS for heart disease might benefit from earlier screening and lifestyle interventions[103]. Furthermore, as diverse biobanks continue to grow, the insights gained will help develop interventions that are effective across diverse populations. Expanding biobanks to integrate multi-omics (such as transcriptomics and proteomics) will enable a more comprehensive understanding of the biological pathways underlying diseases. Further, integrated omics-based models have also recently shown some promise in risk prediction[104]. However, for precision medicine, more research is warranted on the impact of such models on clinical decision making and health outcomes. Several biobanks are increasingly focusing on implementing comprehensive mechanisms for the return of genomic risk results, to be used by physicians at point-of-care for clinical decision-making[105–108]. Major initiative include Our Future Health in the UK aiming to translate information gained from two decades of expanding GWAS into precision medicine and actionable insights. However, such initiatives, though promising, requires research on the best practices, challenges associated, and especially the potentially impact on personalized health.

In conclusion, biobanks have been instrumental in driving genetic discoveries through GWAS. The wealth of genetic and phenotypic data they provide has led to advances in understanding complex diseases, especially through increased sample sizes, diverse populations, and improved methodologies. As biobanks continue to expand and incorporate innovative technologies, they hold tremendous potential to bring about significant advances in precision medicine and public health.

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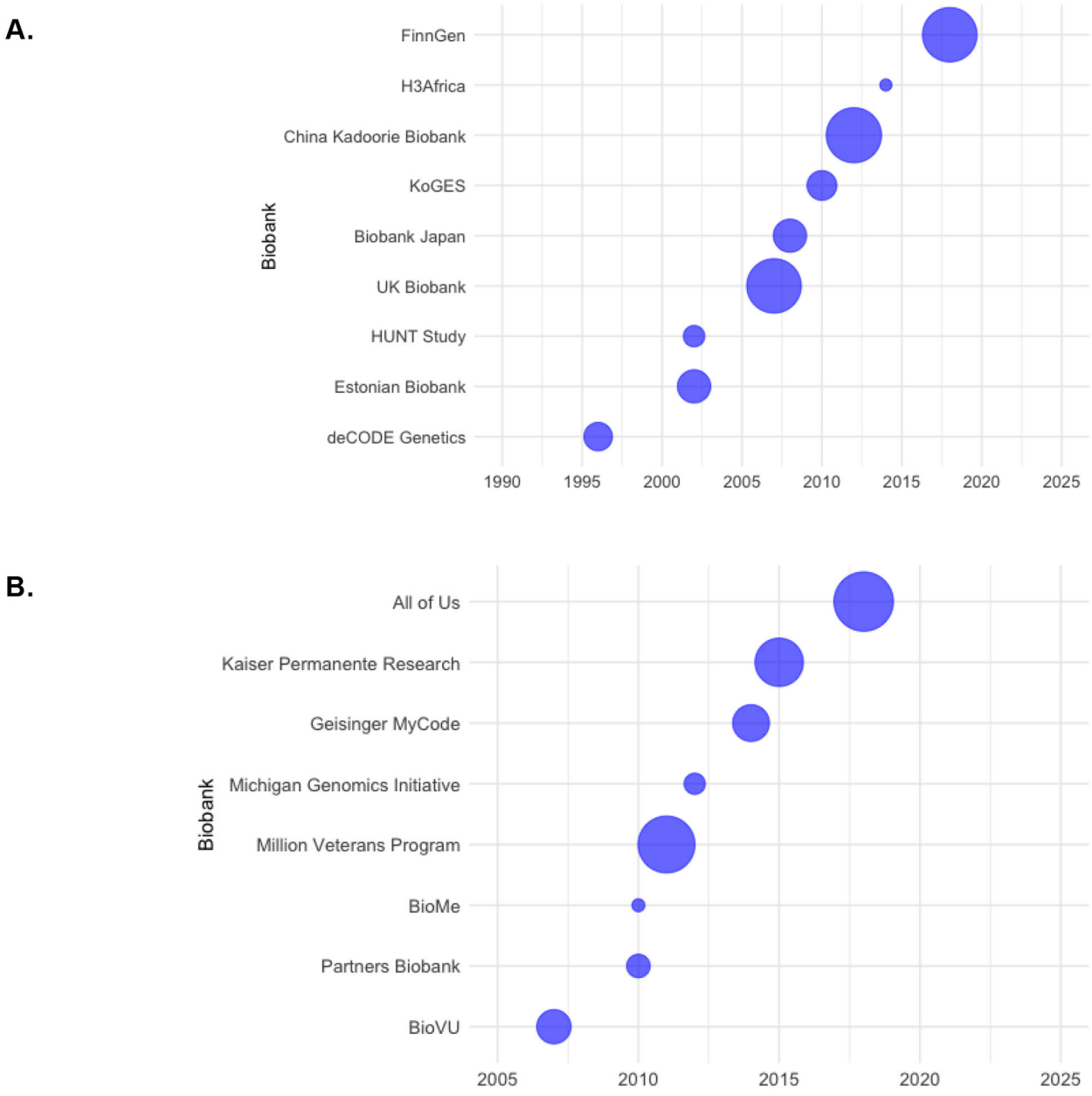


Figure 1.

Evolution and timeline of (A) population based and (B) EHR/health system-based biobanks. For both plots, the biobanks are arranged approximately temporally according to when they started incorporating genetic information. The size of the bubble represents relative sample sizes ranging from 50,000 to 1,000,000. List of biobanks used here do not constitute an exhaustive list.

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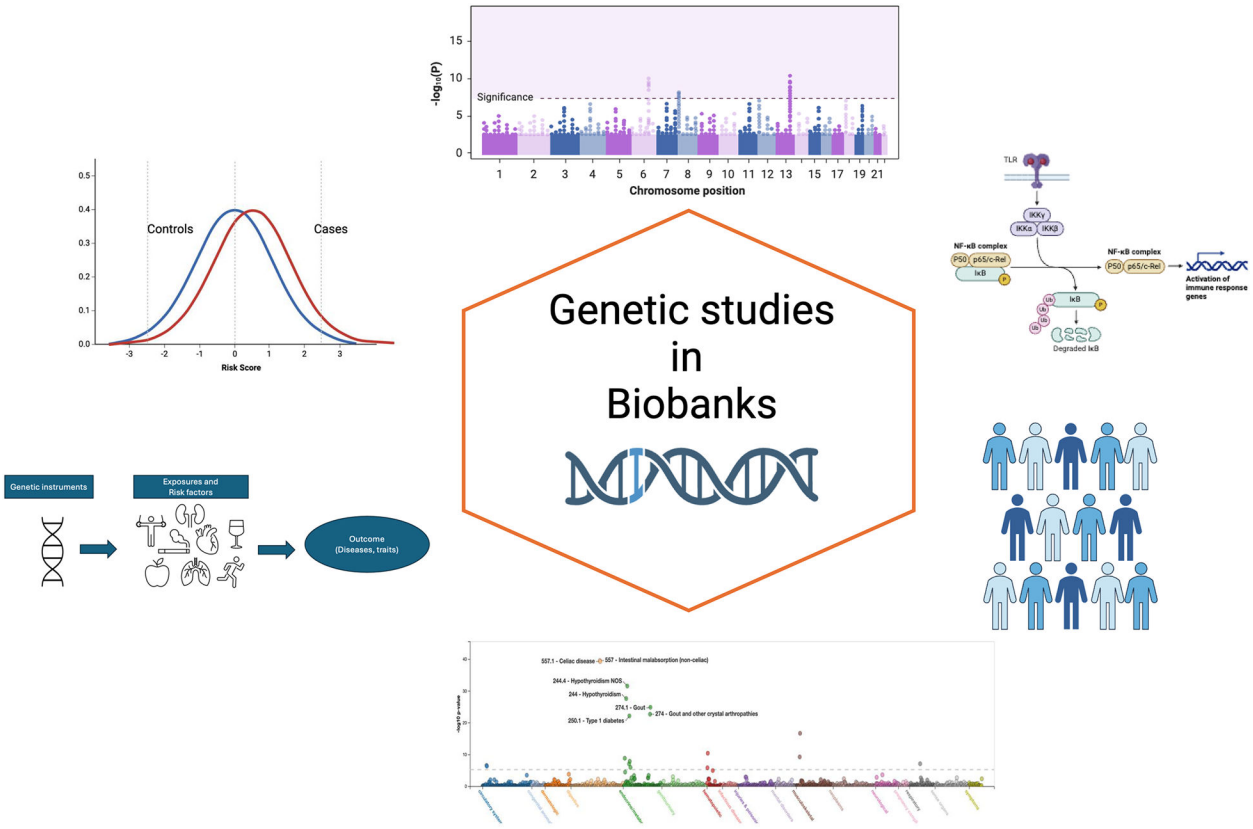


Figure 2.

A graphic illustration of different facets of current genetic studies using biobanks. The graphic was created using Biorender (https://www.biorender.com/)